

<UEE795> : <Capstone Project >				
UEE795: Semester VI(starts) UEE795: SemesterVII (Completion)	L	T	P	Cr
	0	0	2*	8.0
Course Objective: To facilitate the students learn and apply an engineering design process in electrical engineering, including project resource management. As a part of a team, the students will make a project, that emphasizes, hands-on experience, and integrates analytical and design skills. The idea is to provide an opportunity to the students to apply what they have learned throughout the course of graduate program by undertaking a specific problem.				
Syllabus: Capstone Project is increasingly interdisciplinary, and requires students to function on multi-disciplinary teams. It is the process of devising a system, component or process to meet desired needs. It is a decision-making process (often iterative), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs. It typically includes both analysis and synthesis performed in an iterative cycle. Thus, students should experience some iterative design in the curriculum. As part of their design experience, students have an opportunity to define a problem, determine the problem scope and to list design objectives. The project must also demonstrate that students have adequate exposure to design, as defined, in engineering contexts. Engineering standards and realistic constraints are critical in engineering design. The program must clearly demonstrate where standards and constraints are taught and how they are integrated into the design component of the project. Each group will have 4-5 students. Each group should select their team leader and maintain daily diary. Each Group will work under mentorship of a Faculty supervisor. Each group must meet the assigned supervisor (2hrs slot/week) till the end of the semester (record of attendance will be maintained), as per the time slot which will be provided to them by the respective supervisor. This is mandatory requirement for the fulfilment of the attendance as well as the successful completion of the project. The faculty supervisor of the project will continuously assess the progress of the works of the assigned groups. Some part of the analysis and design of the system will be done in the first section of project in semester VI. The second section would comprise of completion of the project in semester VII in which each team will have to submit a detailed report of the project along with a poster.				
Course Learning Outcomes (CLO) After completion of the course, the students will be able to: 1. Identify design goals and analyse possible approaches to meet given specifications with realistic Engineering constraints. 2. Design an electrical engineering project implementing an integrated design approach applying knowledge accrued in various professional courses. 3. Perform simulations and incorporate appropriate adaptations using iterative synthesis. 4. Use modern engineering hardware and software tools. 5. Work amicably as a member of an engineering design team. 6. Improve technical documentation and presentation skills.				

Approved in 109th meeting of the Senate held on March 16,2023. Revised in 112th and 114th meeting of the Senate held on March 11,2023 and March 7,2025, respectively.

SEMESTER-VIII

<UEE 894> : <Project Semester>				
	L	T	P	Cr
	0	0	0	15.0
Course Objective: The project semester is aimed at developing the undergraduate education program in Electrical Engineering to include a practical training in a professional engineering set up (a company, top educational institution, research institute, etc.) hereafter referred to as host “organization” as deemed appropriate. The participating organizations are selected that are either already visiting Thapar Institute of Engineering & Technology(TIET), Patiala for placement or are forming new relationships of mutual benefit. The project semester gives the students, the opportunity to translate engineering theory into practice in a professional engineering environment. The technical activity in the project semester should be related to both the student’s engineering studies and to the host organization s activities and it should constitute as significant body of engineering work at the appropriate level. It should involve tasks and methods that are more appropriately completed in a professional engineering environment and should, where possible, make use of human and technology resources provided by the organization. It consolidates the student’s prior learning and provides a context for later research studies. The student remains a fulltime registered student at TIET, Patiala during the project semester and this activity is therefore wholly distinct from any industrial interactions which may occur over vacation periods.				
Assessment Details: Each student is assigned a faculty supervisor who is responsible for managing and assessment of the project semester. The faculty supervisor monitors the student’s progress in a semester and interacts with the industry mentor during his/her visit to the host organization twice. This includes a Reflective Diary which is updated throughout the project semester, an Interim Project Report, a Final Report with Learning Agreement/Outcomes and a Final Presentation & Viva which involves the faculty Supervisor and some other members from the department. The mentor from the host organization is asked to provide his assessment on the designated form. The faculty supervisor is responsible for managing and performing the assessment of the project semester experience.				
Course Learning Outcomes (CLO) After completion of the course, the students will be able to: 1. Acquire knowledge and experience of software and hardware practices in the area of project. 2. Carry out design calculations and implementations in the area of project. 3. Associate with the implementation of the project requiring individual and teamwork skills. 4. Communicate their work effectively through writing and presentation. 5. Demonstrate the knowledge of professional responsibilities and respect for ethics.				